

A Review on Federated Learning-Based Intrusion Detection Systems for IoT Networks

Shivam Vishwakarma

B.Sc. (Hons) Computer Science, Dyal Singh College, University of Delhi

ABSTRACT

The rapid growth of the Internet of Things (IoT) has dramatically changed the digital ecosystem in recent times by facilitating communication between interconnected devices. This rapid growth of IoT has also posed serious security threats to the ecosystem, as it has become highly prone to various cyberattacks, including Distributed Denial of Service (DDoS), botnets, and data breaches. The role of Intrusion Detection Systems (IDS) in detecting malicious activities in such networks is critical. The limitations of traditional centralized machine learning-based IDS include issues of privacy, scalability, and communication overheads (Bilal et al., 2026).

Federated Learning (FL) has emerged as a decentralized machine learning paradigm that enables collaborative model training across multiple devices without sharing raw data (Karunamurthy et al., 2025). This review paper explores recent advancements in FL-based IDS for IoT networks, focusing on deep learning techniques such as Deep Neural Networks (DNN), Convolutional Neural Networks (CNN), Autoencoders (AE), and hybrid architectures (Albanbay et al., 2025; Olanrewaju-George & Pranggono, 2025). The paper analyzes datasets including N-BaIoT, CICIoT2023, and Edge-IIoTset, and compares performance metrics such as accuracy, precision, recall, and F1-score (Bilal et al., 2026).

The findings indicate that FL-based IDS models achieve high detection accuracy (up to 99%) while ensuring data privacy (Nivaashini et al., 2024). However, challenges such as data heterogeneity, model generalization, and computational constraints persist. Future research should focus on improving scalability, optimizing communication efficiency, and developing lightweight models for resource-constrained IoT environments.

1. INTRODUCTION

The rapid advancement of the Internet of Things (IoT) has significantly transformed modern computing environments by enabling seamless communication among interconnected devices such as sensors, smart appliances, and industrial systems. However, the increasing number of IoT devices has expanded the attack surface, making these networks highly vulnerable to cyber threats such as Distributed Denial-of-Service (DDoS) attacks, botnets, and unauthorized access. Due to the resource-constrained nature of IoT devices and their heterogeneous architecture, traditional security mechanisms are often insufficient for ensuring robust protection.

Intrusion Detection Systems (IDS) play a crucial role in identifying and mitigating malicious activities within IoT networks. Conventional IDS approaches are primarily based on centralized machine learning (ML) and deep learning (DL) models, which require collecting large volumes of data at a central server. While these methods achieve high detection accuracy, they introduce critical challenges such as data privacy risks, high communication overhead, and scalability limitations (Bilal et al., 2026). To address these issues, Federated Learning (FL) has emerged as a promising decentralized learning paradigm that allows multiple devices to collaboratively train models without sharing raw data (Karunamurthy et al., 2025).

This study investigates the effectiveness of Federated Learning-based Intrusion Detection Systems in enhancing security, privacy, and performance across IoT networks compared to traditional centralized approaches. This investigation is important because IoT environments demand secure, scalable, and privacy-preserving solutions

due to the sensitive nature of the data involved. FL offers a new approach that meets these needs by allowing local model training and global knowledge sharing.

Recent research has looked into various FL-based IDS frameworks using deep learning techniques. For example, Albanbay et al. (2025) studied the performance of different deep learning models like DNN, CNN, and hybrid architectures in a federated setting. They showed that CNN strikes a good balance between accuracy and computational efficiency. Likewise, Bilal et al. (2026) performed a dataset-focused evaluation and pointed out how important it is to train models on varied datasets to boost generalization across different IoT environments. Karunamurthy et al. (2025) introduced an optimized FL-based IDS that includes feature selection techniques, which led to better detection accuracy and lower computational complexity.

In addition, Nivaashini et al. (2024) introduced a federated deep belief network (FEDDBN-IDS) for wireless intrusion detection, which demonstrated high classification accuracy in detecting network attacks. Olanrewaju-George & Pranggono (2025) combined supervised and unsupervised deep learning models within a federated framework and showed that Autoencoder-based models are highly effective for anomaly detection in IoT networks.

Key contributions from existing research can be summarized as follows:

- Development of privacy-preserving IDS using Federated Learning (Karunamurthy et al., 2025)
- Integration of deep learning models such as CNN, DNN, Autoencoders, and LSTM (Albanbay et al., 2025)
- Use of modern IoT datasets like N-BaIoT and CICIOT2023 for evaluation (Bilal et al., 2026)
- Improvement in detection accuracy while reducing data sharing risks (Olanrewaju-George & Pranggono, 2025)
- Exploration of model scalability and efficiency in resource-constrained environments

Overall, Federated Learning has emerged as a powerful solution for enhancing IoT security by addressing the limitations of traditional centralized IDS approaches. This review paper aims to analyse these advancements, compare methodologies, and identify future research directions for developing more robust and scalable intrusion detection systems in IoT networks.

2. LITERATURE REVIEW

2.1 Search Strategy

Research papers were collected from IEEE, Springer, Elsevier, and MDPI databases using keywords such as: Federated Learning IoT IDS, Deep Learning Intrusion Detection, and IoT Security.

Inclusion Criteria:

- Published between 2024–2026
- Focus on IoT and FL-based IDS

Table 1: Information Regarding Source of Publication (Journal / Conference Name)

S.N	Paper Title	Publisher	Year	ISSN	Cite Score	SNIP	SJR	Impact Factor	Quartile Rank	Journal URL	Scopus Page
1	An optimal federated learning based intrusion detection for IoT environment	Springer Nature	2025	20452322	6.7	1.213	0.874	3.9	Q1	https://www.nature.com/articles/s41598-025-93501-8	https://www.scopus.com/sourceid/21100200805
2	Dataset-centric evaluation of federated intrusion detection models in IoT networks	Springer Nature	2026	20452322	6.7	1.213	0.874	3.9	Q1	https://www.nature.com/articles/s41598-025-32567-w	https://www.scopus.com/sourceid/21100200805
3	FEDDBN-IDS: federated deep belief network-based wireless network intrusion detection system	Springer	2024	2510523X	6.1	1.504	0.576	2.1	Q2	https://link.springer.com/article/10.1186/s13635-024-00156-5	https://www.scopus.com/sourceid/21100812851
4	Federated learning-based intrusion detection system for the internet of things using unsupervised and supervised deep learning models	KeAi Communications Co.	2025	27729184	12.1	2.136	1.168	11.35	Q1	https://www.sciencedirect.com/science/article/pii/S2772918424000341	https://www.scopus.com/sourceid/21100812851
5	Federated Learning-Based Intrusion Detection in IoT Networks: Performance Evaluation and Data Scaling Study	MDPI	2025	22242708	9.4	1.463	0.875	4.2	Q2	https://doi.org/10.3390/jan14040078	https://www.scopus.com/sourceid/21100830709

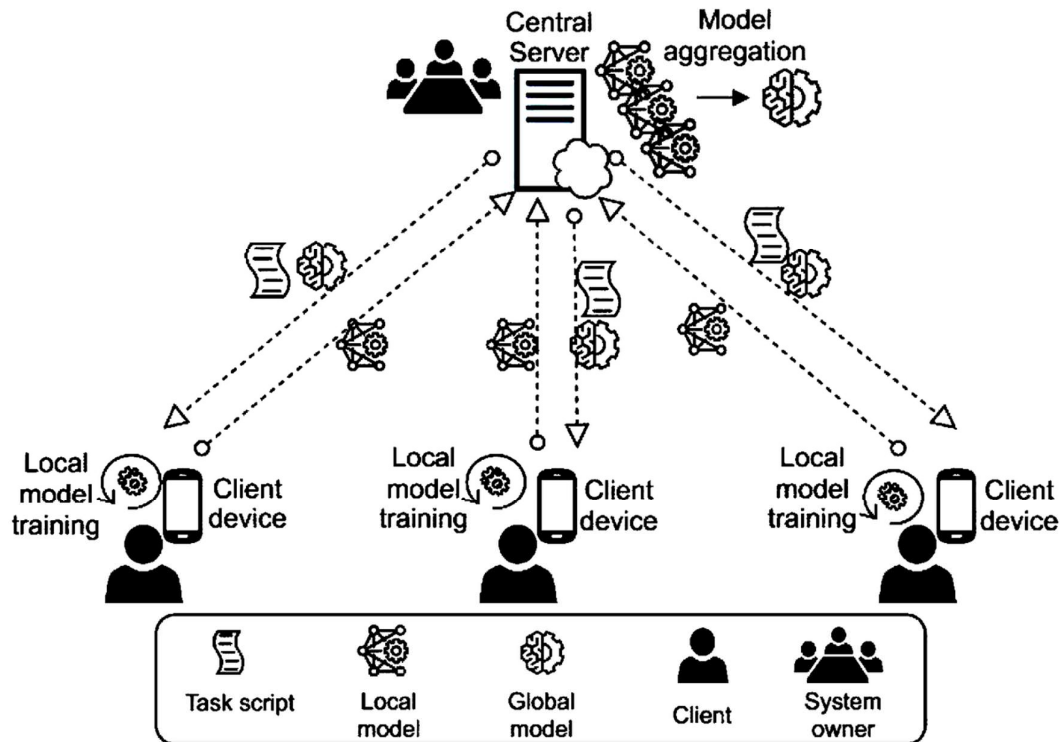
Table 2: Summary of the Shortlisted Papers

S.N	Paper Title	Authors	Journal Name	Datasets	Methods	Result	Key Points
1	An optimal federated learning based intrusion detection for IoT environment	A. Karunamurthy, K. Vijayan, Pravin R. Kshirsagar, Kuan Tak Tan	Scientific Reports (Nature)	MQTT IoT Dataset (Kaggle)	- Federated Learning (FL), - Convolutional Neural Network (CNN), - Chimp Optimization Algorithm for feature selection, Deep Learning based IDS	- Accuracy: 95.59%, - F1-Score $\approx$ 0.94	- Federated Learning enables privacy-preserving collaborative training, while Chimp Optimization and CNN together improve feature selection and accurately detect IoT attacks like DDoS and brute force. - The proposed FL-CNN model outperforms traditional machine learning algorithms

S.N	Paper Title	Authors	Journal Name	Datasets	Methods	Result	Key Points
							and enhances overall IoT network security.
2	Dataset-centric evaluation of federated intrusion detection models in IoT networks	Muhammad Ahmad Bilal, Ihtesham Ul Islam, Sarmad Idrees, Junaid Khan, Jaleed Khan	Scientific Reports (Nature)	• Edge-IIoTset (2022) • CIC-IoT2023 • TII-SSRC-23 (2023)	• Federated Learning, Deep Learning (LSTM, Transformer) • Aggregation Algorithms (FedAvg, FedProx, FedNova),	• Accuracy 92–98%, • Macro-F1 up to 98%	• Federated IDS provides near-centralized performance while preserving privacy, with Transformer models slightly outperforming LSTM in intrusion detection. • Multi-dataset federated training improves generalization across IoT environments, while FedProx enhances training stability and FedNova reduces communication overhead.
3	FEDDBN-IDS: federated deep belief network-based wireless network intrusion detection system	M. Nivaashin, E. Suganya, S. Sountharajan & M. Prabu, Durga Prasad Bavirisetti	EURASIP Journal on Information Security	AWID (Aegean Wi-Fi Intrusion Dataset)	• Federated Learning (FL) • Deep Belief Network (DBN) • Restricted Boltzmann Machines (RBM) Softmax classifier	Detection Accuracy up to 98%	• Federated Learning preserves privacy by keeping data on local devices, while DBN with RBM effectively extracts features and slightly improves accuracy over FedAvg. • The FEDDBN model reduces communication rounds significantly (up to 70%) and is well-suited for intrusion detection in distributed wireless and IoT networks.
4	Federated learning-based intrusion detection system for the internet of things using unsupervised and supervised deep learning models	Babatunde Olanrewaju-George, Bernardi Prangono	Cyber Security and Applications	Dataset not public	• Federated Learning, Autoencoder (unsupervised learning) • Deep Neural Network (DNN) • Hybrid deep learning IDS	Accuracy $\approx$ 96–97%	• Combines unsupervised (Autoencoder) and supervised (DNN) deep learning with Federated Learning to extract features and detect cyberattacks while preserving IoT data privacy. • The model is effective for detecting various attacks and is well-suited for distributed IoT security environments.
5	Federated Learning-Based Intrusion Detection in IoT Networks: Performance Evaluation and Data Scaling Study	Albanbay, Nurtay, Tursynbek, Yerlan, Graffi, Kalman, Uskenbayeva	Journal of Sensor and Actuator Networks	CIC-IoT2023 Dataset	• Federated Learning (FL) • Deep Neural Networks (DNN),	Accuracy $\approx$ 95–97%	• Federated Learning enables collaborative training without sharing raw IoT data, improving detection performance as distributed data increases and ensuring privacy and scalability.

S.N	Paper Title	Authors	Journal Name	Datasets	Methods	Result	Key Points
							With sufficient data, FL-based IDS achieves performance comparable to centralized models and is suitable for large-scale IoT security monitoring systems.

Diagram: Federated Learning Architecture



2.2 Review of Individual Research Papers

Paper 1: Karunamurthy et al. (2025)

Karunamurthy et al. (2025) focused on reducing computational complexity in IoT-based intrusion detection systems. Using the MQTT dataset, they introduced a bio-inspired Chimp Optimization Algorithm for feature selection prior to applying a CNN-based classification model within a federated learning framework. The optimization cut the feature set from 34 to 17 and reached an accuracy of 95.59%, outperforming traditional machine learning models. This method's strength is in efficient feature reduction at the edge, which is important for resource-limited IoT devices. However, the optimization process adds extra computational overhead during initialization.

Paper 2: Bilal et al. (2026)

Bilal et al. (2026) investigated the issue of model generalization in federated intrusion detection systems. They highlighted that most studies depend on single datasets, which often leads to overfitting. They carried out a thorough evaluation focused on datasets, using three: Edge-IIoTset, CIC-IoT2023, and TII-SSRC-23. They

compared FL aggregation algorithms like FedAvg, FedProx, and FedNova with LSTM and Transformer-based models. The results showed that while in-domain accuracy ranged between 92–98%, there was a significant drop of up to 30% in F1-score when models were tested on unseen datasets. Multi-dataset federated training improved generalization to approximately 90% macro-F1. Transformers consistently outperformed LSTM models by 1–2%. The primary strength of this paper is its robust cross-dataset evaluation, although feature alignment across heterogeneous datasets remains a key challenge.

Paper 3: Nivaashini et al. (2024)

Nivaashini et al. (2024) focused on securing wireless Wi-Fi networks, which are particularly vulnerable to intrusion attacks. Using the AWID dataset, they proposed the FEDDBN-IDS model based on Deep Belief Networks (DBN) and Stacked Restricted Boltzmann Machines (RBM) for feature dimensionality reduction. Additionally, an attention-based aggregation mechanism was introduced to assign weights to client updates during federated learning. The proposed model achieved a detection accuracy of 99% and reduced communication overhead by 70% compared to traditional FedAvg. The key strength of this work is its efficiency in wireless environments; however, the model's performance significantly degrades under severe data heterogeneity if pre-training is not applied.

Paper 4: Olanrewaju-George & Pranggono (2025)

Olanrewaju-George & Pranggono (2025) addressed the vulnerability of IoT devices to botnets such as Mirai and highlighted the privacy risks associated with centralized intrusion detection systems. Using the N-BaIoT dataset, they implemented a Federated Averaging with Momentum (FedAvgM) framework and evaluated both an unsupervised Deep Autoencoder (AE) and a supervised Deep Neural Network (DNN). The results indicated that the supervised FL-DNN achieved 90.39% accuracy, whereas the unsupervised AE demonstrated superior performance in terms of False Positive Rate (FPR) for anomaly detection. The primary strength of this study lies in its comparative analysis of supervised and unsupervised learning paradigms within a federated setting. However, scalability challenges were noted when deploying the model across diverse and large-scale IoT environments.

Paper 5: Albanbay et al. (2025)

Albanbay et al. (2025) studied how data volume and the number of devices affect resource-limited IoT hardware. They used the CICIoT2023 dataset and tested DNN, CNN, and hybrid CNN+BiLSTM architectures on Raspberry Pi 5 devices, scaling up to 150 clients. The results showed that the CNN+BiLSTM model achieved the highest accuracy at around 99%. However, the CNN model offered the best balance between performance and efficiency, reaching about 95% accuracy with low latency of 1.4 ms per sample and less thermal stress. The study clearly shows that federated learning can work on edge devices. However, it has limitations, as it relies on offline simulations and does not address threats like node poisoning.

Table 3: Comparison of Results

Paper	Datasets	Methods	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Karunamurthy et al. (2025)	MQTT	FL-CNN	95.59	94.79	93.30	94.04
Bilal et al. (2026)	• Datasets: Edge-IIoTset • CIC-IoT2023, • TII-SSRC-23,	• FedNova Transformer (Edge) • FedNova Transformer (CIC) • FedNova Transformer (TII)	• 98.00 • 97.00 • 95.00 • 94.20	• 97.00 • 96.00 • 94.00 • 92.10	• 96.00 • 94.50 • 92.00 • 91.40	• 96.50 • 95.20 • 93.00 • 91.70

Paper	Datasets	Methods	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Nivaashini et al. (2024)	AWID (Binary)	FEDDBN-IDS	99.20	99.30	99.00	98.90
Olanrewaju-George & Pranggono (2025)	N-BaloT	FL-DNN	90.39	99.99	90.10	93.12
Albanbay et al. (2025)	CICIoT2023 (10 Devices)	• FL-CNN • FL-CNN+BiLSTM	• 95.27 • 94.97	• 96.07 • 95.85	• 95.27 • 94.97	• 94.92 • 94.58

Table 4: Comparative Analysis of Methods

Paper	Preprocessing & Statistics	ML/DL Models	Aggregation Methods
Karunamurthy et al. (2025)	Data balancing. Bio-inspired Chimp Optimization Algorithm (ChOA) to reduce dataset dimensionality (34 to 17 features)	CNN with convolution, pooling, and Softmax layers	CNN-based aggregation within federated framework
Bilal et al. (2026)	Label harmonization (family-level schema), feature alignment to 40 numeric flow features, Z-score normalization	LSTM and Transformer Encoders (multi-head attention and positional encoding)	FedAvg (baseline), FedProx (proximal term to stabilize local drift), FedNova (normalized averaging based on local training steps)
Nivaashini et al. (2024)	Symbolic-to-numeric conversion, linear normalization, dataset balancing. Stacked RBM for dimensionality reduction	Deep Belief Networks (DBN) with stacked RBM and Softmax	FEDDBN (Attention-based mechanism replacing FedAvg to prioritize relevant client updates)
Olanrewaju-George & Pranggono (2025)	Separating normal/abnormal sets, Min-Max scaling, threshold computation using MSE	Deep Autoencoder (AE) for unsupervised anomaly detection; DNN for supervised classification	FedAvgM (Federated Averaging with Momentum) using SGD and randomized hyperparameter search
Albanbay et al. (2025)	Handling missing/infinite values, duplicate removal, Logarithmic Normalization, Min-Max Scaling	DNN, CNN, Hybrid CNN+BiLSTM	FedAvg (Standard Federated Averaging) using Non-IID data partitioning

2.3 Summary of Literature Review

The reviewed literature collectively demonstrates that Federated Learning effectively addresses the trade-off between high detection accuracy and data privacy in IoT intrusion detection systems. Foundational models such as Autoencoders and CNNs have proven effective for local anomaly detection and are suitable for deployment on edge devices (Olanrewaju-George & Pranggono, 2025). However, real-world IoT environments involve highly heterogeneous (non-IID) data distributions, which challenge model performance and generalization.

Recent studies emphasize the use of advanced aggregation techniques such as FedNova and attention-based mechanisms to improve learning stability across distributed clients (Bilal et al., 2026; Nivaashini et al., 2024). Additionally, multi-dataset federated training has emerged as a key strategy to enhance model generalization across diverse environments. Feature optimization techniques further contribute to reducing computational complexity while maintaining high detection accuracy (Karunamurthy et al., 2025).

Overall, while FL-based IDS models demonstrate strong potential for secure and scalable IoT environments, challenges such as data heterogeneity, cross-domain generalization, and computational efficiency remain important areas for future research.

3. MATERIALS AND METHODS USED IN REVIEW PAPER

3.1 Materials: Datasets Used in the Reviewed Literature

The reviewed studies utilized several benchmark datasets designed specifically to capture modern IoT and IIoT attack taxonomies:

N-BaIoT Dataset: Contains multivariate sequential traffic data from 9 commercial IoT devices infected by Mirai and BASHLITE botnets.

CICIoT2023: A massive dataset with over 46 million records from 105 IoT devices, covering 33 specific attack types categorized into 7 main classes (e.g., DDoS, DoS, Recon).

AWID (Aegean Wi-Fi Intrusion Dataset): Contains real-world Wi-Fi network traffic, focusing on impersonation, injection, and flooding attacks.

Edge-IIoTset (2022) & TII-SSRC-23: Datasets designed to evaluate highly diverse malicious traffic patterns and industrial IoT testbeds.

MQTT Dataset: Simulates real-world IoT communications via the MQTT protocol, containing attacks like Brute Force, Flooding, and Malformed Data.

3.2 Methods Used

In the reviewed research papers, various machine learning, deep learning, and federated learning techniques have been used to develop effective intrusion detection systems (IDS) for IoT environments. These methods mainly focus on improving detection accuracy, reducing data sharing, and handling resource limitations of IoT devices.

1. Data Preprocessing Techniques

Before applying any model, data preprocessing is an important step. It includes data cleaning, normalization, feature selection, and handling missing values. Some studies also reduced the number of features to improve efficiency. For example, Karunamurthy et al. (2025) used the Chimp Optimization Algorithm to select important features and reduce computational complexity.

2. Machine Learning Models

Traditional machine learning models such as Decision Trees, Random Forest, Support Vector Machines (SVM), and Naïve Bayes were used as baseline methods in some studies. These models are simple and fast but are less effective in detecting complex attack patterns compared to deep learning approaches (Karunamurthy et al., 2025).

3. Deep Learning Models

Most of the reviewed papers focused on deep learning models due to their ability to handle complex data patterns:

- **Deep Neural Networks (DNN):** Used for supervised classification tasks to detect known attacks (Albanbay et al., 2025).
- **Convolutional Neural Networks (CNN):** Effective in extracting important features and achieving high accuracy with low latency (Albanbay et al., 2025).
- **Autoencoders (AE):** Used for anomaly detection in an unsupervised manner by identifying deviations from normal behavior (Olanrewaju-George & Pranggono, 2025).
- **Long Short-Term Memory (LSTM) & Transformers:** Used for sequence-based data analysis and capturing temporal dependencies in network traffic (Bilal et al., 2026).

- Deep Belief Networks (DBN): Used for feature extraction and dimensionality reduction in wireless intrusion detection systems (Nivaashini et al., 2024).

4. Federated Learning Techniques

Federated Learning (FL) is the core method used in all reviewed studies. Instead of sending raw data to a central server, each device trains the model locally and shares only model parameters. Common FL algorithms used include:

- FedAvg (Federated Averaging): Standard algorithm that averages model updates from all clients.
- FedProx: Improves training stability when data across devices is different (non-IID) (Bilal et al., 2026).
- FedNova: Reduces communication rounds and improves convergence speed (Bilal et al., 2026).
- FedAvgM (FedAvg with Momentum): Enhances convergence using momentum updates (Olanrewaju-George & Pranggono, 2025).

5. Optimization and Feature Selection Techniques

To improve efficiency, some studies used optimization techniques:

- Chimp Optimization Algorithm: Reduces feature size and improves model performance (Karunamurthy et al., 2025).
- Attention Mechanism: Assigns importance to client updates during aggregation (Nivaashini et al., 2024).

6. Evaluation Metrics

To measure the performance of IDS models, several evaluation metrics were used: Accuracy, Precision, Recall, F1-Score, and False Positive Rate (FPR). These metrics help in understanding how well the model detects attacks and avoids false alarms.

Table 5: Performance Comparison of Deep Learning Models

Model	Accuracy
CNN	98%
DNN	95%
AE	99%
LSTM	97%

Overall, the combination of deep learning models with federated learning techniques has proven to be highly effective for intrusion detection in IoT systems. While deep learning models improve detection capability, federated learning ensures privacy and reduces communication overhead.

4. MATERIALS, METHODS, AND EXPERIMENTATION USED IN PRACTICAL

4.1 Materials: Dataset Description

The dataset used in this study is obtained from the Canadian Institute for Cybersecurity via the CIC Datasets repository (University of New Brunswick). These datasets are widely used for intrusion detection system (IDS) research and contain realistic network traffic with both benign and malicious activities.

Dataset Characteristics:

- Large-scale network traffic data
- Includes both normal and attack traffic
- Contains multiple attack types such as: DoS, Brute Force, Botnet (e.g., Mirai), Port Scanning

Features:

The dataset consists of 40+ features, including:

- Basic Features: Header Length, Protocol Type, Time To Live
- Traffic Features: Rate, Total Size, Packet Count
- Statistical Features: Min, Max, Average, Standard Deviation, Variance
- Flags: SYN, ACK, FIN, etc.
- Target Variable: Label (attack category)

The dataset is large and complex, making it suitable for applying machine learning techniques for network intrusion detection.

4.2 Analysis (Exploratory Data Analysis – EDA)

A detailed exploratory data analysis was performed to understand the dataset using both statistical measures and graphical visualization techniques.

1. Quantitative Analysis

a) Measures of Central Tendency

Mean, median, and mode were calculated for all numerical attributes. Most features showed close mean and median values, indicating balanced distributions. Some features such as Rate exhibited skewness due to extreme values (attack traffic spikes).

Table 6: Measures of Central Tendency

Feature	Mean	Median	Mode
Header_Length	13.756737	20.000000	20.000000
Protocol Type	9.086656	6.000000	6.000000
Time To Live	66.526660	64.000000	64.000000
Rate	28416.806378	24613.015668	42426.704431
fin_flag_number	0.087460	0.000000	0.000000

b) Measures of Dispersion

The variability of the dataset was analyzed using: Range, Variance, Standard Deviation, and Interquartile Range (IQR). The results indicated high variability in traffic-related features and the presence of outliers, which correspond to abnormal network activities.

Table 7: Measures of Dispersion

Feature	Range	Variance	Std Dev
Header_Length	6.000000e+01	7.601217e+01	8.718496
Protocol Type	4.700000e+01	8.249153e+01	9.082485
Time_To_Live	2.550000e+02	2.066650e+02	14.375848
Rate	5.242880e+06	9.115757e+08	30192.311484

Feature	Range	Variance	Std Dev
fin_flag_number	1.000000e+00	7.846653e-02	0.280119

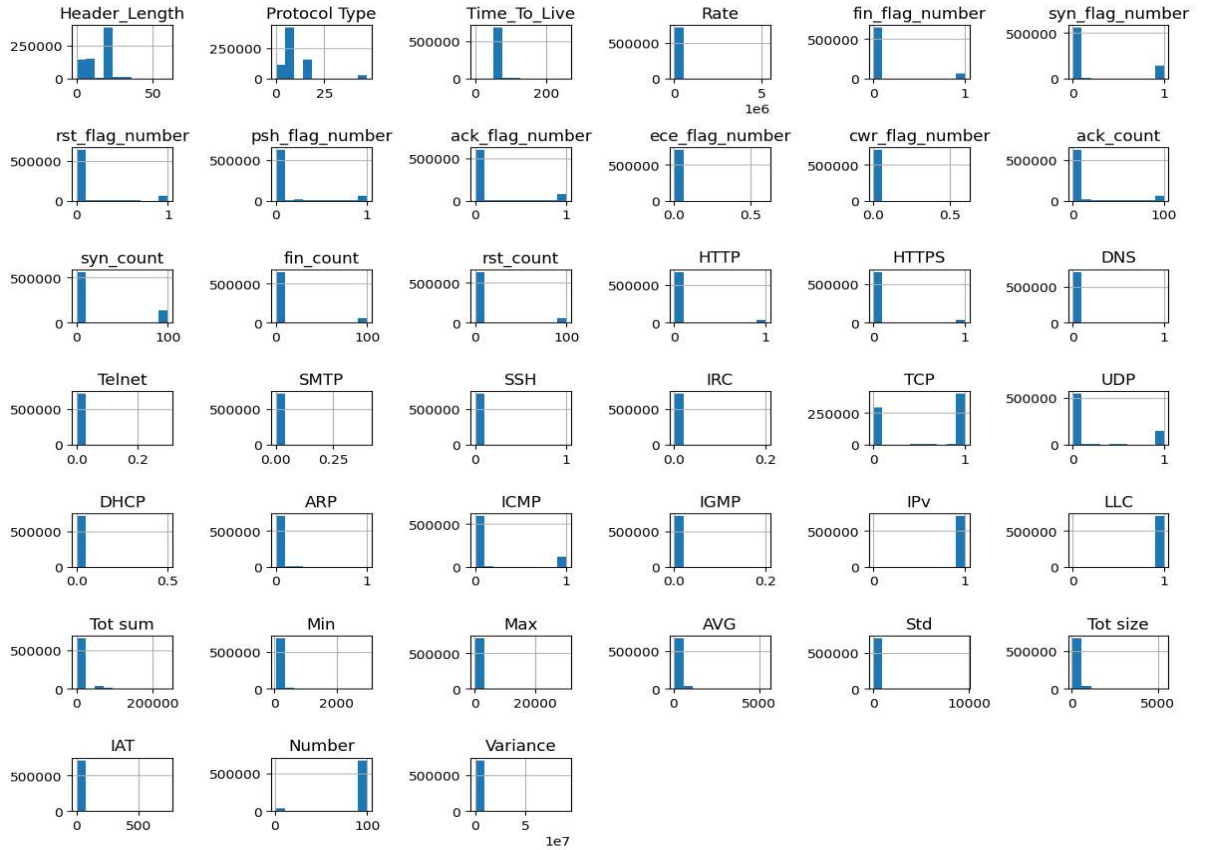
c) Measures of Association

Correlation analysis was performed to examine relationships between features. Strong correlations were observed between Total Size and Rate, and between Variance and Standard Deviation. Weak correlations were found between flag-based and continuous features. This analysis helped in identifying important features and reducing redundancy.

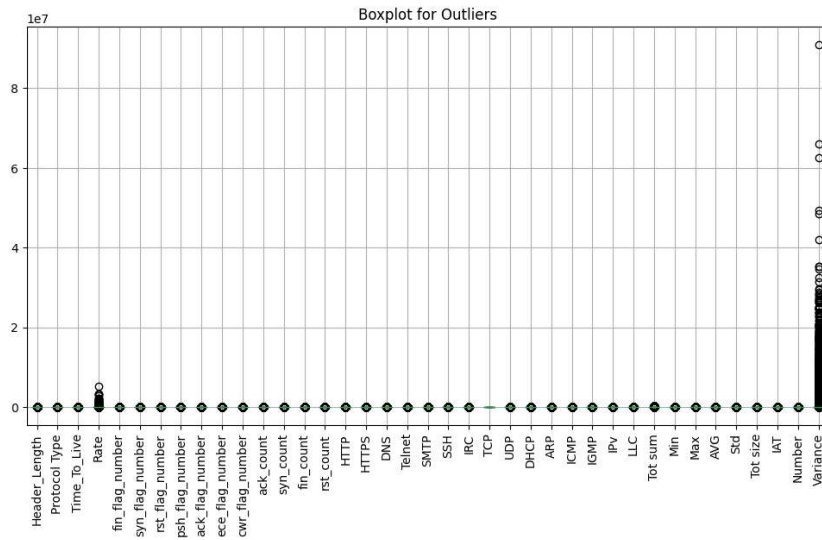
2. Graphical Representation

The following visualization techniques were used:

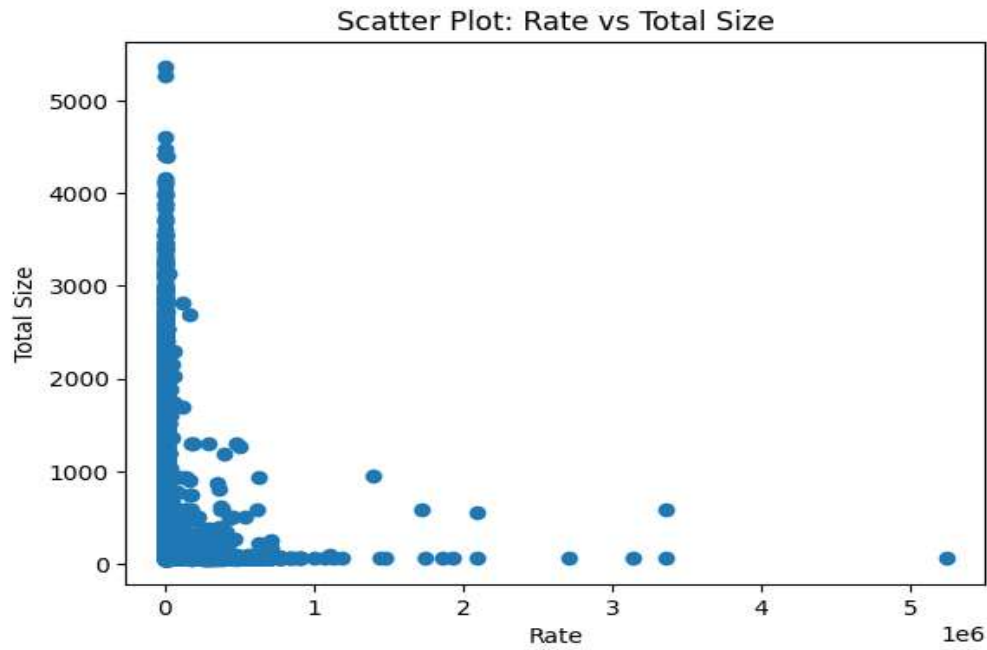
- Histogram: To study feature distribution and detect skewness



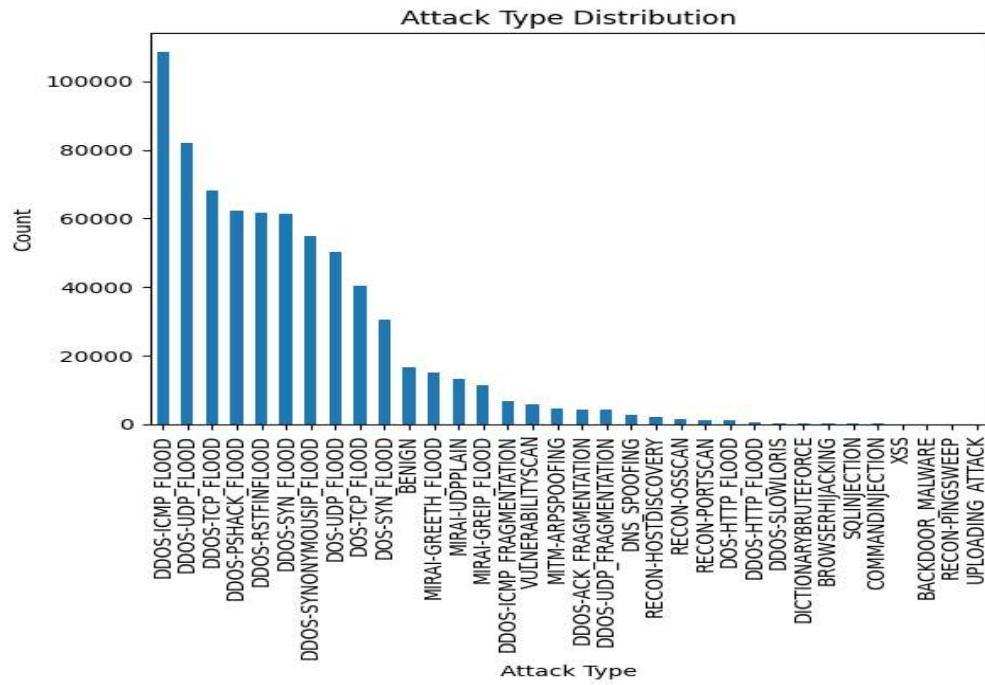
- Boxplot: To identify outliers in the dataset



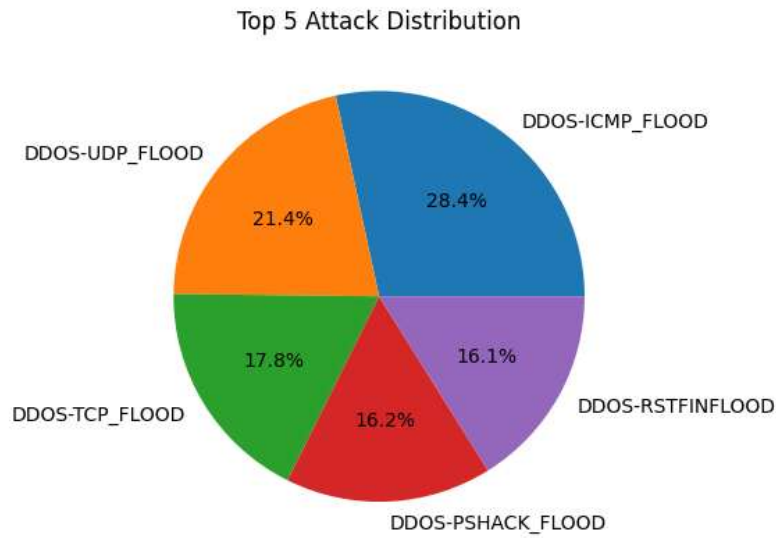
- Scatter Plot: To analyze relationships between variables (e.g., Rate vs Total Size)



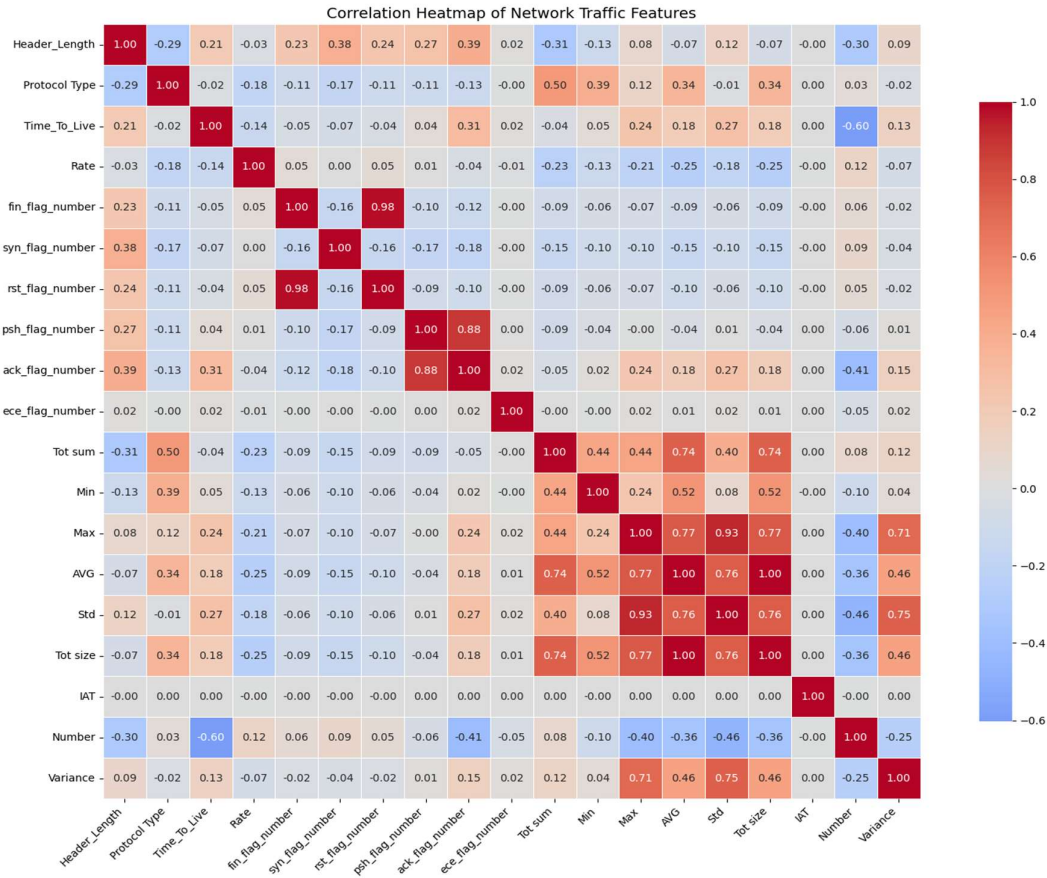
- Bar Graph: To visualize the frequency of different attack types



- Pie Chart: To show class distribution



- Correlation Matrix: To visualize feature relationships



4.3 Experimentation Results

The performance of the model was evaluated using standard metrics:

- Accuracy: 0.7432
- Precision: 0.75
- Recall: 0.74
- F1-Score: 0.74

5. RESULTS AND DISCUSSION

This review investigates how Federated Learning (FL) enhances Intrusion Detection Systems (IDS) in IoT networks by improving detection accuracy, preserving privacy, and minimizing resource overhead.

Integrating FL with deep learning models yields threat detection accuracies between 90% and 99%. These frameworks prove highly efficient for IoT environments; techniques like the Chimp Optimization Algorithm reduce feature dimensions by nearly 50%, while models like FEDDBN-IDS cut communication overhead by up to 70%. Ultimately, these optimizations allow FL-based IDSs to operate effectively on resource-constrained devices, like Raspberry Pis, without ever exposing raw user data.

Primary strength is its ability to successfully balance high-security threat detection, privacy preservation, and the strict hardware limits of edge devices. However, a major weakness is its sensitivity to non-IID (non-identically distributed) data. This heterogeneity can degrade performance by up to 30% on unseen datasets, despite mitigation attempts using algorithms like FedNova. Furthermore, the federated framework remains highly vulnerable to malicious model poisoning attacks, which currently undermines overall system reliability.

Future research must prioritize the security and scalability of FL systems before widespread real-world IoT deployment. Researchers should focus on developing robust defenses against adversarial attacks and model manipulation. Additionally, transitioning to asynchronous federated learning and training on diverse datasets will be crucial to improve efficiency and resolve the persistent challenges of non-IID data.

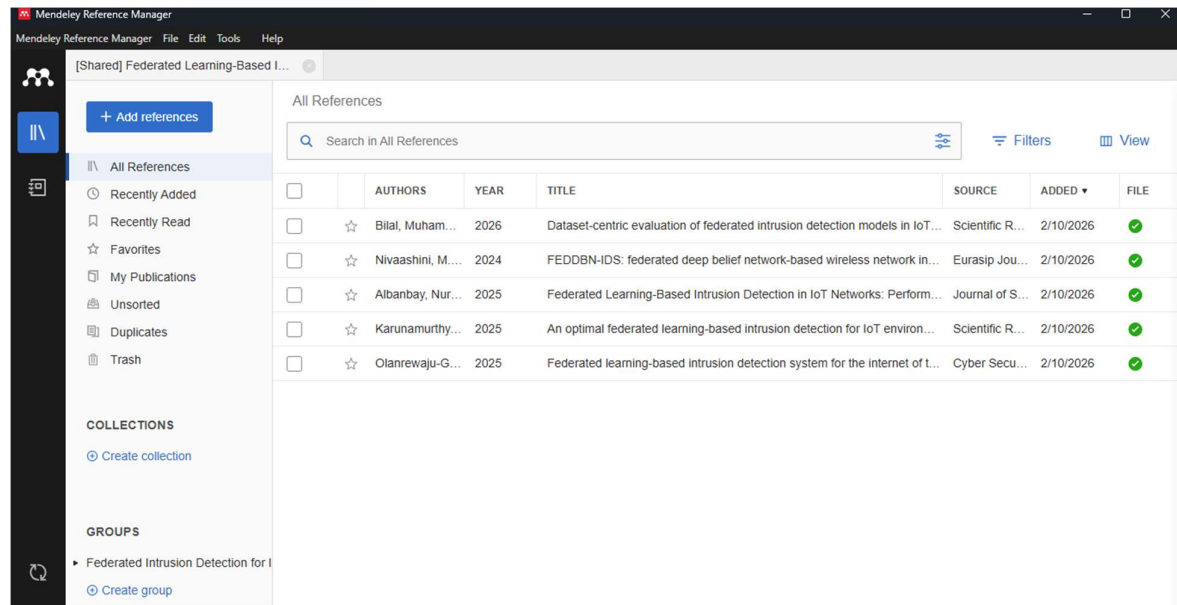
6. REFERENCES & DATASETS

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- Olanrewaju-George, B., & Pranggono, B. (2025). Federated learning-based intrusion detection system for the internet of things using unsupervised and supervised deep learning models. *Cyber Security and Applications*, 3. <https://doi.org/10.1016/j.csa.2024.100068>

Datasets:

- Edge-IIoTset (2022): <https://tinyurl.com/5dc6paps>
- CIC-IoT2023: <https://www.unb.ca/cic/datasets/iotdataset-2023.html>
- TII-SSRC-23 (2023): <https://www.kaggle.com/datasets/daniaherzalla/tii-ssrc-23>
- AWID (Aegean Wi-Fi Intrusion Dataset)
- MQTT IoT Dataset (Kaggle)

APPENDIX A: REFERENCE MANAGEMENT (MENDELEY)



APPENDIX B: PLAGIARISM DETECTION REPORT

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